

Synchrotron emission -

$$P = \frac{e^2 a^2}{6\pi\epsilon_0 c^3} \rightarrow a = \gamma v \times B \sin\theta \quad \text{pitch angle}$$

$$\bullet \theta = 0 \rightarrow \parallel \text{ to } \vec{B}, P = 0 \propto \sin^2\theta$$

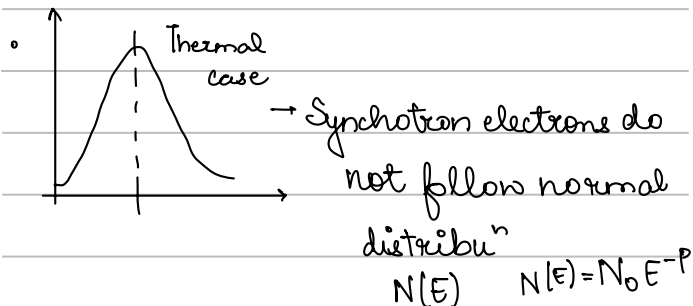
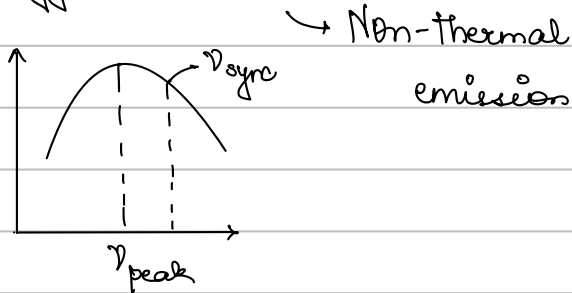
$$\bullet \theta = \pi/2 \rightarrow \text{max. power}$$

$$0 \leq \theta \leq 90^\circ \rightarrow v \approx c$$

$$P = \frac{e^2}{6\pi\epsilon_0 c^3} \times \frac{\gamma^2 v^2 B^2 \sin^2\theta}{m_e^2} = \frac{e^4}{6\pi\epsilon_0 c} \frac{\gamma^2 B^2 \sin^2\theta}{m_e^2}$$

$$E = \gamma m_e c^2 \rightarrow \frac{e^4}{6\pi\epsilon_0 c} \frac{E^2}{m_e^2 c^4} \frac{B^2 \sin^2\theta}{m_e^2} \propto E^2 B^2$$

$\bullet N(E)dE \rightarrow$ Total no. of electrons within the energy bin E to $E+dE$



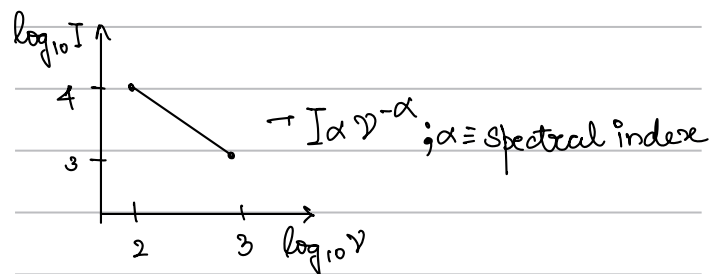
$\bullet$ Power-law distribution $\rightarrow$

$$N(E) \propto E^{-p} \quad \text{III}$$

$\bullet 10 \text{ keV } e^- \rightarrow$ lose energy faster than 1 keV but becomes quicker faster also

$\bullet$ Intensity $\propto \gamma^{-\alpha}$

$10,000 \text{ Jy} \rightarrow 100 \text{ MHz}$
 $1000 \text{ Jy} \rightarrow 1 \text{ GHz}$ } not following RJ, had to be non-thermal

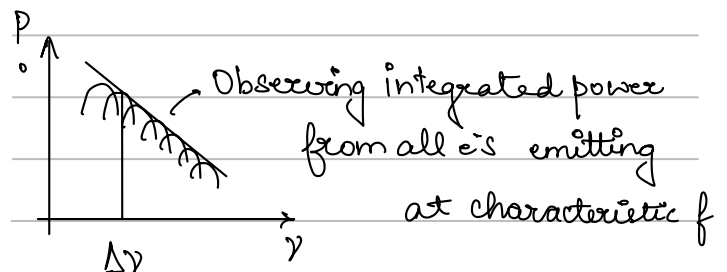


$$\bullet \gamma \propto E^2 B$$

$$N(E)dE = N_0 E^{-p} dE$$

$$\bullet P \propto I \Rightarrow P \propto \gamma^{-\alpha}$$

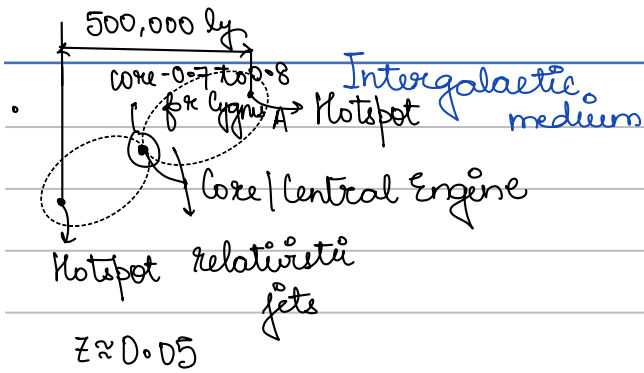
$$\hookrightarrow (E^2)^{-\alpha}$$



$$P_{\text{tot}} = \int P_\gamma \cdot d\gamma$$

$$\bullet E^{2\alpha} \cdot 2E dE \propto E^{2-p} dE$$

$$\alpha = \frac{1-p}{2} \rightarrow \text{Popula}^n \text{ of electrons}$$



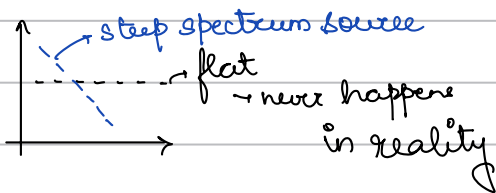
↓ we have very good measurements at
 $2' \times 4'$ diff. frequencies & diff. morphologies

- IGM boundary → lesser effects
- Termination shock

• Telescope resolⁿ less than $2' \times 4'$, I would just see or measure the integrated flux

• $I \propto S^\alpha \rightarrow \alpha \approx -0.7$ to -0.8

When $\alpha = 0$, I is same at all frequencies - flat spectrum source



- Populaⁿ of e⁻s causing emission change in real time
- Fermi acceleraⁿ
- Breakover frequency - unique for every source